

Cohort Analysis Project Description

Description

A comprehensive data analysis project focused on evaluating user retention rates (Retention Rate) through cohort analysis methodology. The project combines SQL database operations with Google Sheets visualization to compare retention patterns between users acquired through promotional campaigns versus organic channels.

My Tasks

Database Query & Data Preparation (SQL)

- Connect to database and explore raw tables (cohort_users_raw, cohort_events_raw)
- Parse and clean dates in multiple formats using SQL regex and CASE statements
- Create CTEs with normalized timestamp fields
- Join user and event tables with proper filtering (remove NULL values, test events)
- Calculate cohort months and user tenure (month_offset)
- Aggregate data into final cohort table with user counts
- Export results as CSV file

Cohort Table Construction (Google Sheets)

- Import SQL results into Google Sheets
- Build pivot table with cohort months as rows, month_offset as columns, user counts as values
- Create second table calculating Retention Rate (%) for each cohort
- Apply conditional gradient formatting to highlight trends
- Implement interactive Slicer for promo_signup_flag filtering

Comparative Analysis & Insights

- Analyze retention patterns between promotional and organic users
 - Document findings and prepare conclusions
 - Create summary sheet with business insights
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Result

✓ Working SQL script that:

- Parses dates in multiple formats (day-month-year with various delimiters)
- Applies proper date validation and transformation
- Joins tables correctly with appropriate filters
- Produces aggregated cohort table in specified format

✓ Interactive Google Sheets dashboard featuring:

- Three well-labeled sheets: Data, Cohort_tables, Conclusions
- Cohort table showing user counts by acquisition month and tenure
- Retention Rate table (%) with proper calculations
- Functional Slicer for filtering promo vs. organic users
- Professional conditional formatting with gradient fill
- Empty cells properly preserved in both tables

✓ Business Insights:

- Clear comparison of retention behavior between promotional and organic user cohorts
 - Assessment of acquisition quality by channel
 - Conclusions about user engagement patterns across different onboarding methods
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Skills Acquired

- Cohort Analysis methodology - identifying user acquisition cohorts and tracking lifecycle metrics
 - SQL proficiency - date parsing with regex, complex JOINS, aggregation functions, CASE statements
 - Data transformation - cleaning inconsistent date formats and normalizing raw data
 - Google Sheets advanced features - Pivot Tables, Slicers, conditional formatting
 - Retention Rate calculation - measuring user engagement over time
 - Data visualization - presenting analytics for stakeholder decision-making
 - Comparative analysis - evaluating performance across different user acquisition channels
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